Z-motion correction

validation & benchmarking

optimouse 2-photon pipeline — C57_O1M2/10022023

CaImAn 1.12.2 · Suite2p 0.14.0 · real 41-plane z-stack

1. End-to-end pipeline validation

Validation of the current method on real anatomy

The z-motion correction chain was validated on real data (C57_O1M2/10022023):

mcorr → z-motion (F_anat) → ROI extraction → per-ROI fit-and-subtract.

Coverage:

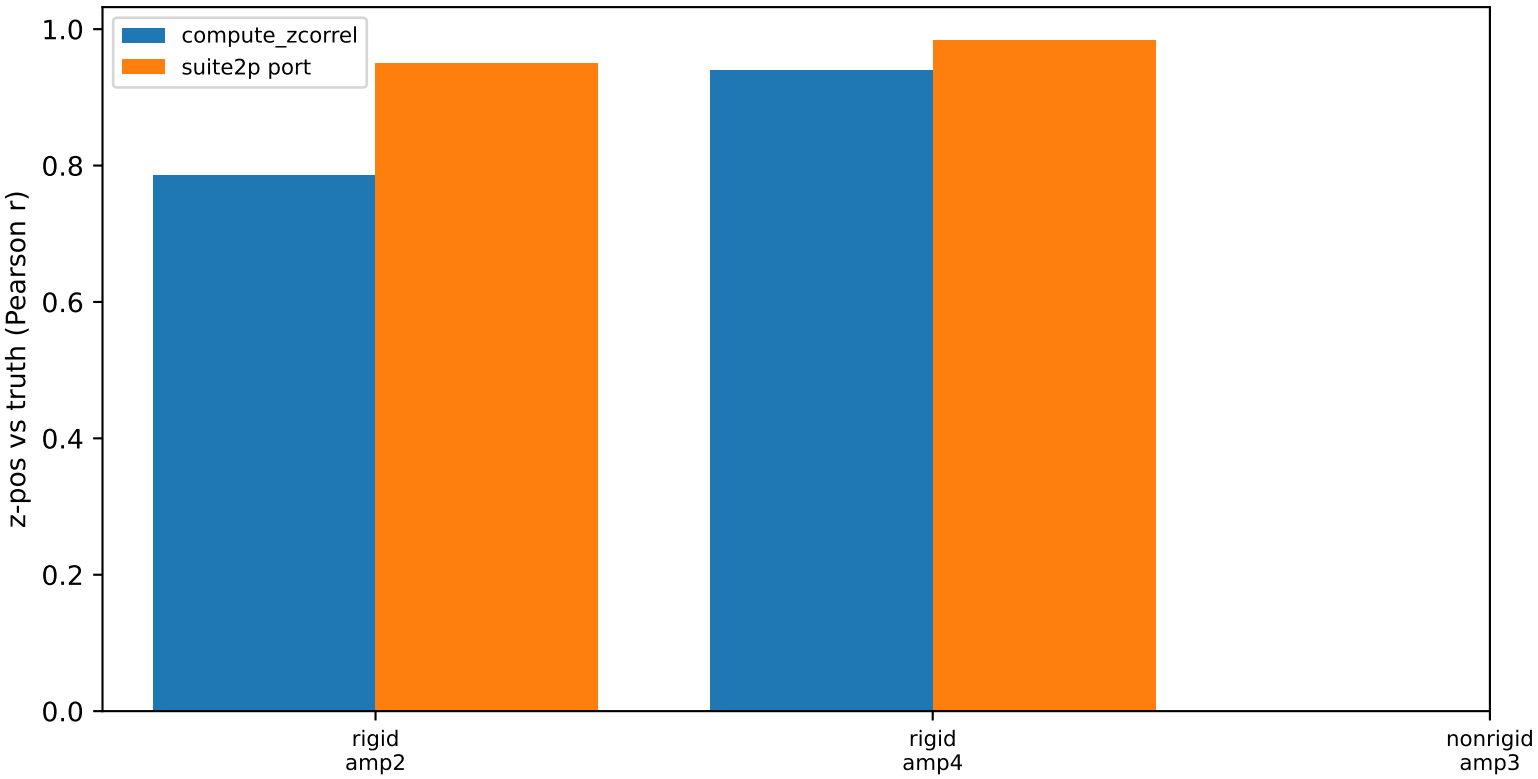
- Subset (TSeries-001, 200 frames) and full session (1600 frames).
- Both extraction backends: CNMF and Suite2p (0.14.0).
- Full session: real z-drift over planes 9–16 (~7 µm, σ≈1 µm); 593 ROIs → F_roi_zcorrected.

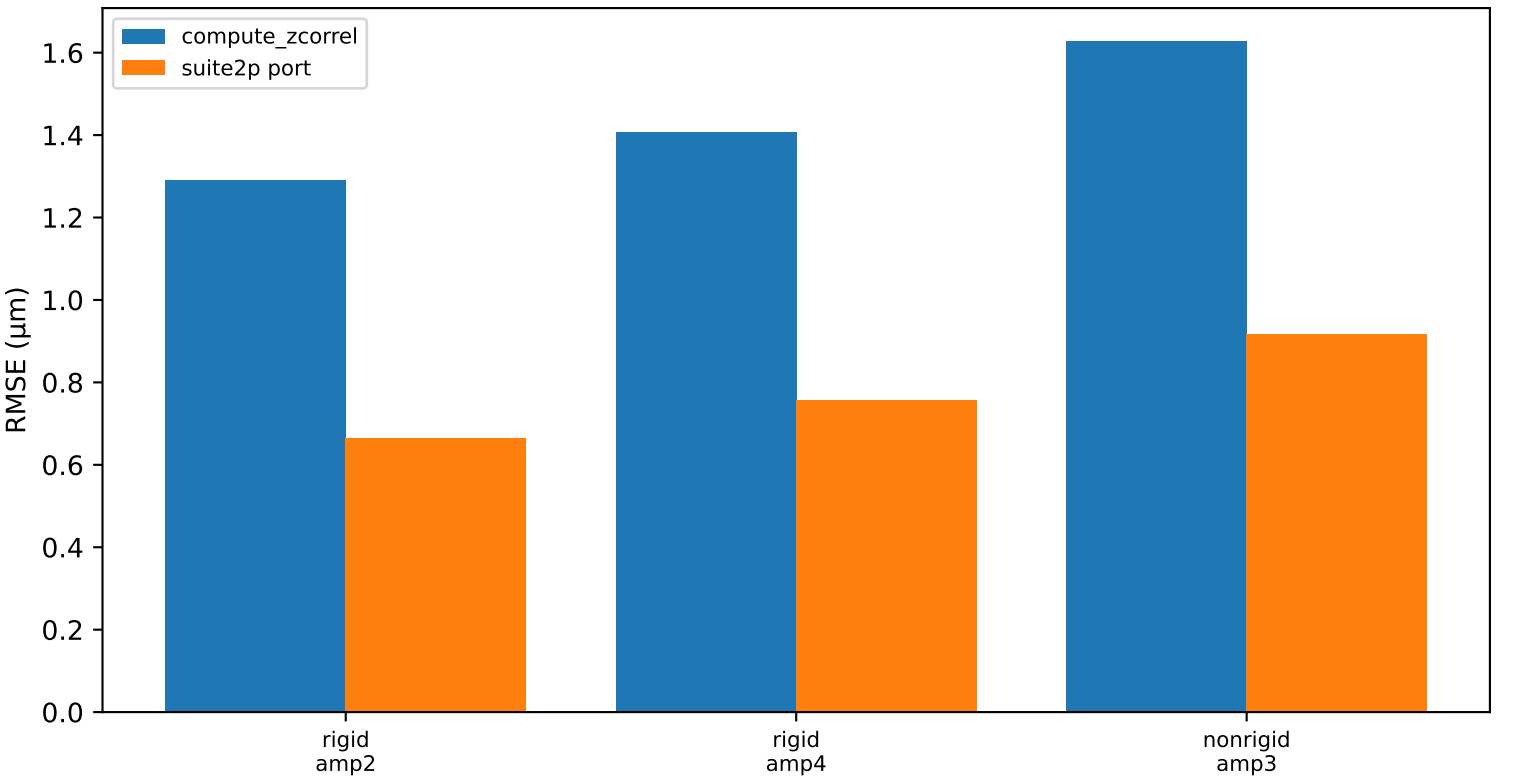
7 bugs fixed in the under-tested per-ROI path (normally done in MATLAB):

- create_components_movie excerpt overflow on short recordings.
- CNMF extraction now writes cnmf_result.hdf5 for run_roi_zcorr.
- load_CNMF(n_processes=1); raw F = C+YrA; dims from cnm.dims (caiman 1.12).
- compute_zcorrel_suite2p: contiguous flip + worker-scope rigid (suite2p 0.14).

Environment: pixi (CaImAn 1.12.2 + Suite2p 0.14.0 pinned to production). The pipeline requires an activated env — bare `python` may be Python 2.

2. z-position — synthetic method sweep (accuracy vs drift amplitude)





Suite2p is more accurate (higher r, lower RMSE) but slower — time/case: rigid amp2: 13.8s vs 82.6s rigid amp4: 2.6s vs 7.9s nonrigid amp3: 2.8s vs 8.0s

3. z-position: Pearson (shipped) vs Suite2p

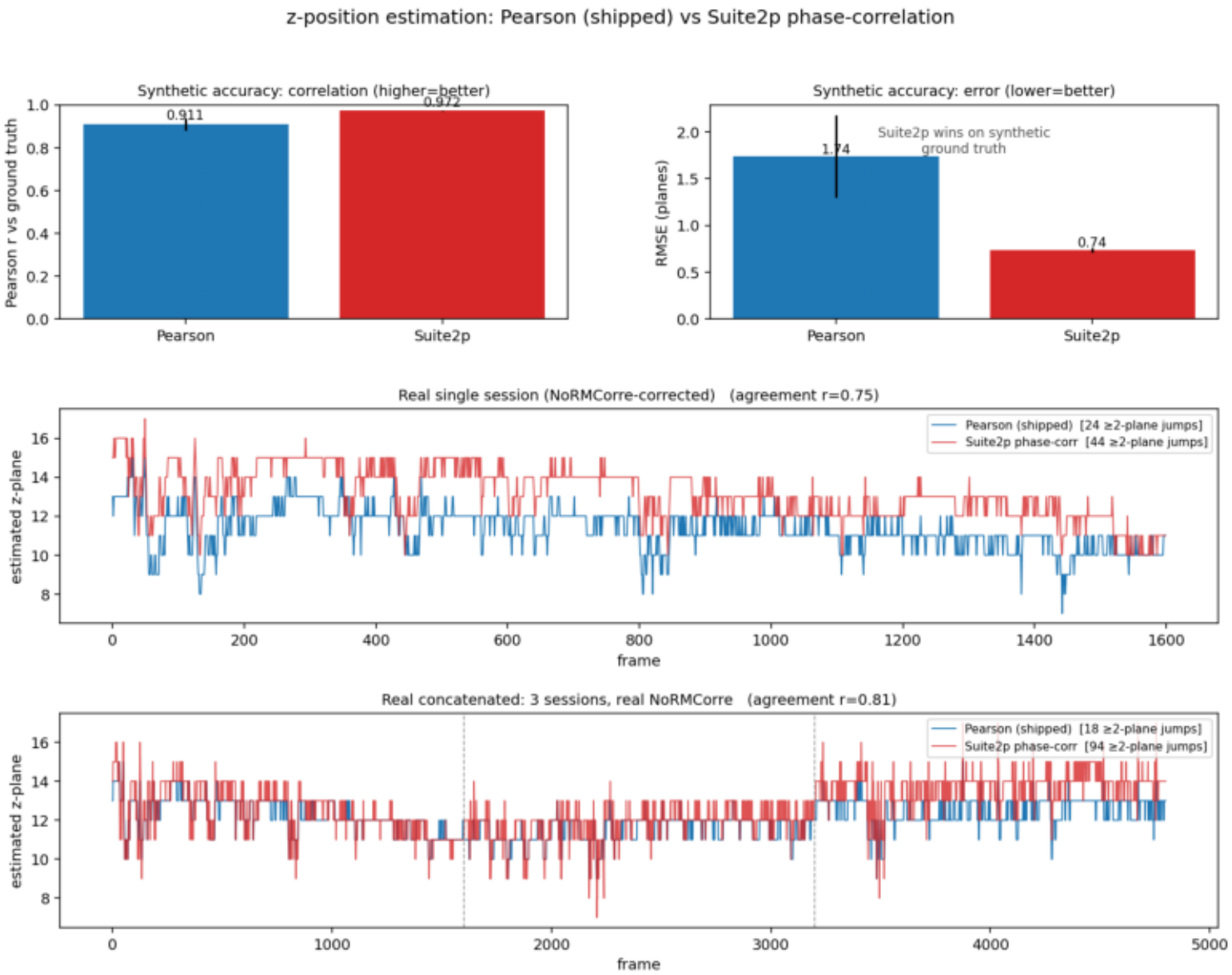
Head-to-head of the two z-position estimators: the shipped Pearson full-frame correlation (+ temporal median smoothing) vs Suite2p masked phase-correlation.

A. On SYNTHETIC ground truth Suite2p is more accurate (r 0.972 vs 0.911, RMSE 0.74 vs 1.74 planes).

B/C. On REAL NoRMCorre output the ranking reverses: phase-correlation is much noisier (and biased toward deep, low-contrast planes in single sessions), because mcorr edge artifacts (border_nan="copy") corrupt the masked correlation. Pearson + temporal smoothing tracks a stable trajectory across sessions (dashed = session boundaries).

On the 3-session real-NoRMCorre concat (C), phase-correlation makes 94 spurious ≥2-plane jumps vs Pearson's 18 (5× more) while tracking the same overall drift (agreement r=0.81).

Since z-position is estimated AFTER motion correction in production, the real-data behaviour is what matters → the pipeline ships Pearson (no Suite2p dependency).



4. z-position — optimized compute_zcorrel (legacy → smoothed)

compute_zcorrel_legacy = full-frame Pearson (kept for comparison).

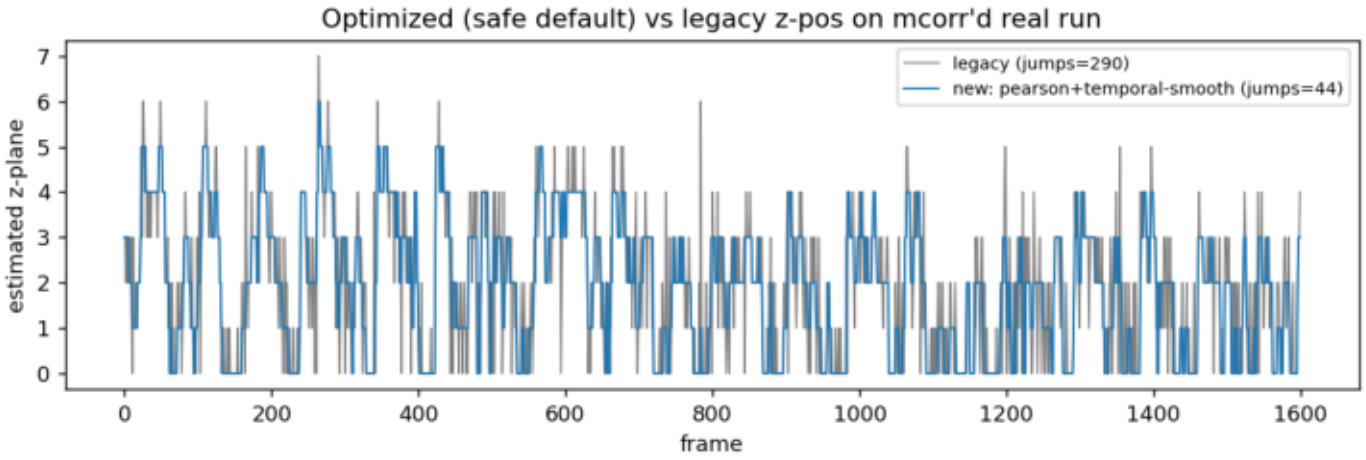
New compute_zcorrel default = Pearson + temporal median smoothing of zpos.

Synthetic ground truth (4 seeds): suite2p/phasecorr is most accurate (RMSE ~0.7 planes) but — see below — fails on mcorr'd data. Pearson+smoothing edges out legacy (r 0.935 vs 0.920).

Two synthetic-motivated options were tested and left OFF by default because they DEGRADE on NoRMCorre-corrected production movies:

- phasecorr (Suite2p): latches onto the deepest low-contrast plane (→ plane 40).
- edge_crop: collapses the Pearson estimate to plane 0.

The one safe win — temporal smoothing — cuts spurious ≥2-plane jumps on the real mcorr'd run from 290 (legacy) to 44, tracking the same drift (r=0.85):

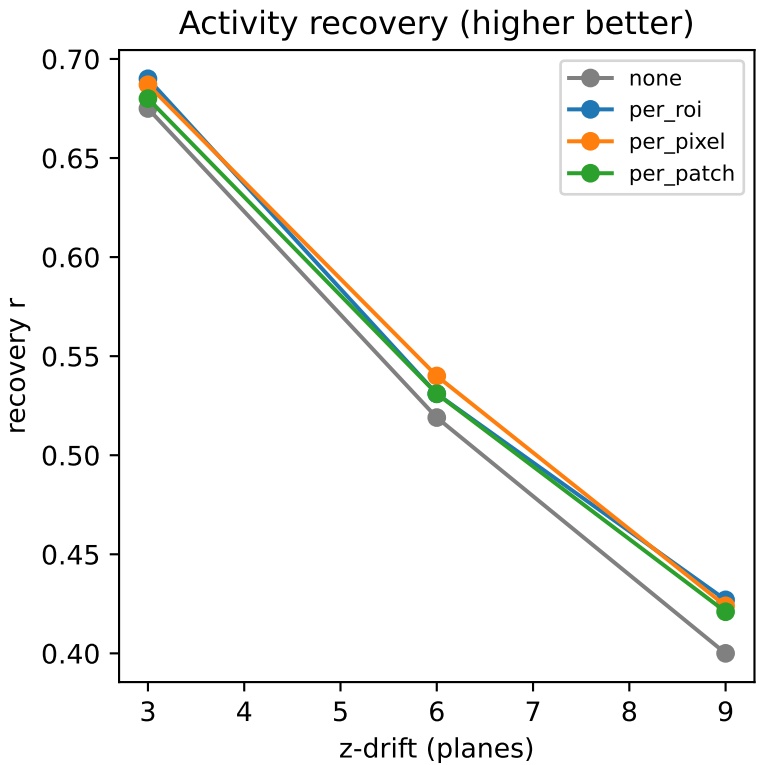


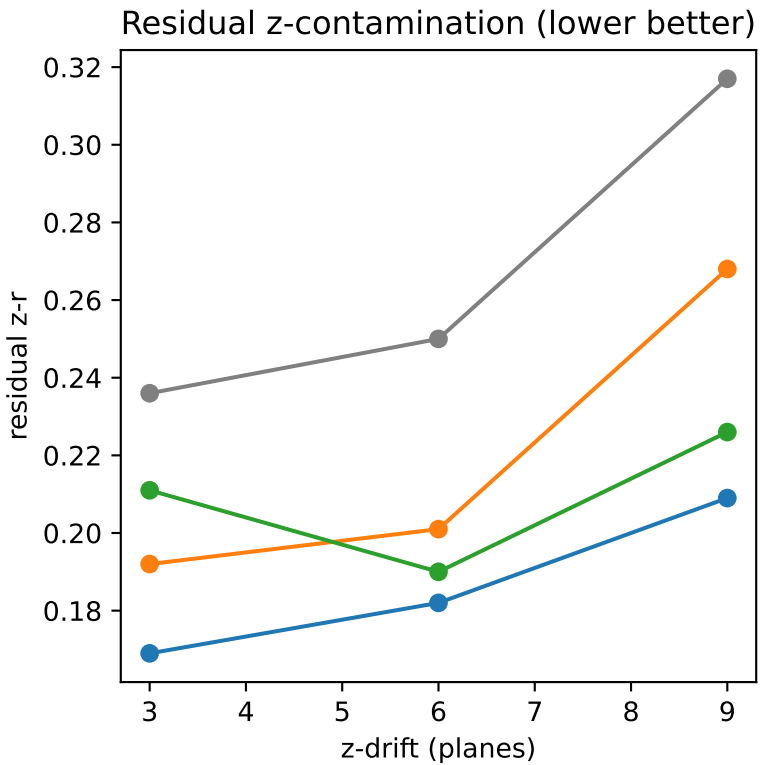
5. z-position — sub-plane interpolation (tested)

case	metric	integer argmax	sub-plane
global z, amp 1	RMSE planes	2.258	2.261
global z, amp 2	RMSE planes	2.790	2.774
global z, amp 4	RMSE planes	1.808	1.791
F_anat corr, amp 3	recovery r	0.721	0.722
F_anat corr, amp 6	recovery r	0.548	0.549

Parabolic interpolation of the correlation peak was expected to beat integer argmax for both the global z-position and the per-patch depth used to build F_anat. Tested: negligible improvement in both. The error is dominated by discrete multi-plane misassignments and defocus signal attenuation, not ±0.5-plane quantization. Conclusion: sub-plane is not worth adding; improve plane-selection robustness or use the more accurate Suite2p estimator instead.

6. Correction — method comparison (recovery vs drift amplitude)





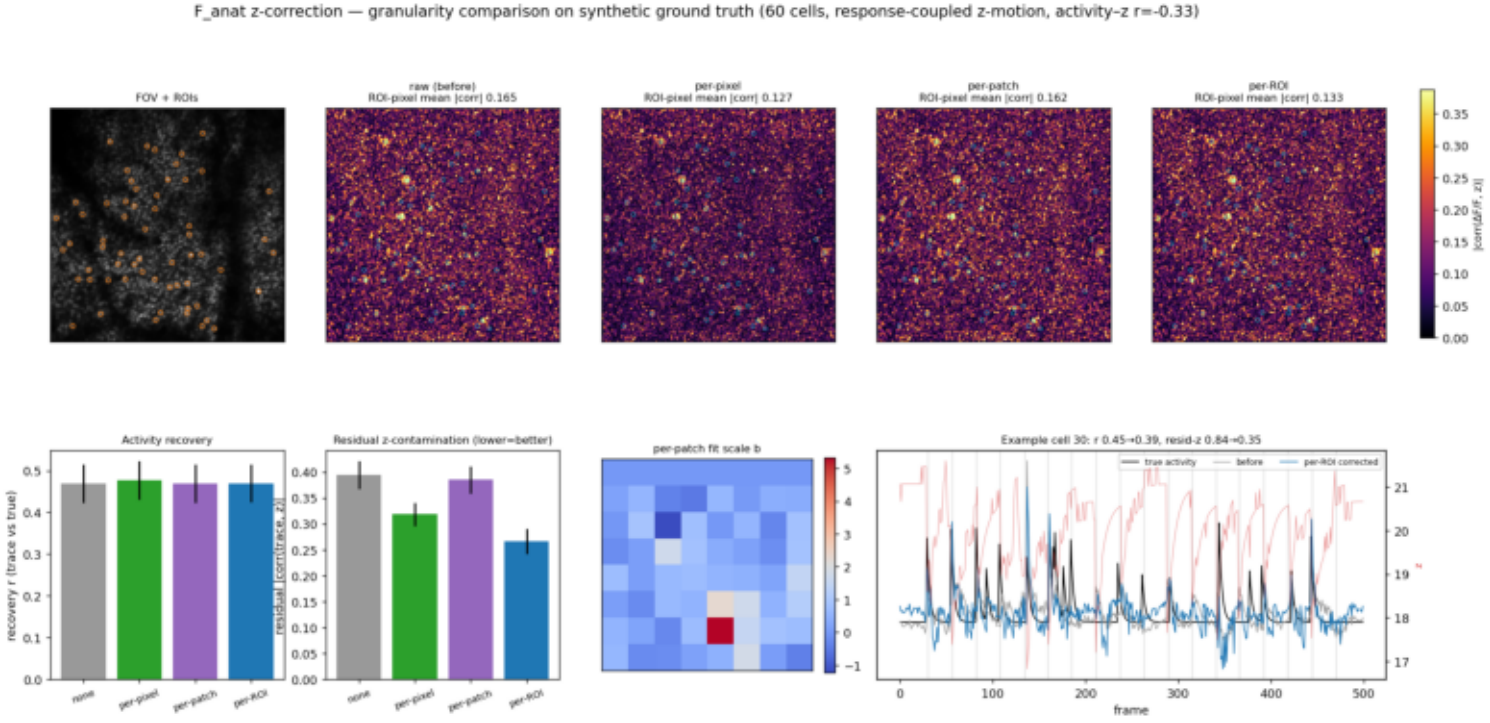
per-ROI removes the most z-contamination at every amplitude; recovery is capped by defocus attenuation.

7. Correction granularity: pixels vs patches vs ROI (synthetic)

The F_anat z-correction can be applied at three spatial granularities. This mirrors the pipeline's patches_summary_plots style: the top row maps per-pixel residual z-contamination |corr(ΔF/F, z)| over the FOV after each method (cyan = ROI centers); the bottom row quantifies recovery and residual z on the extracted ROI traces.

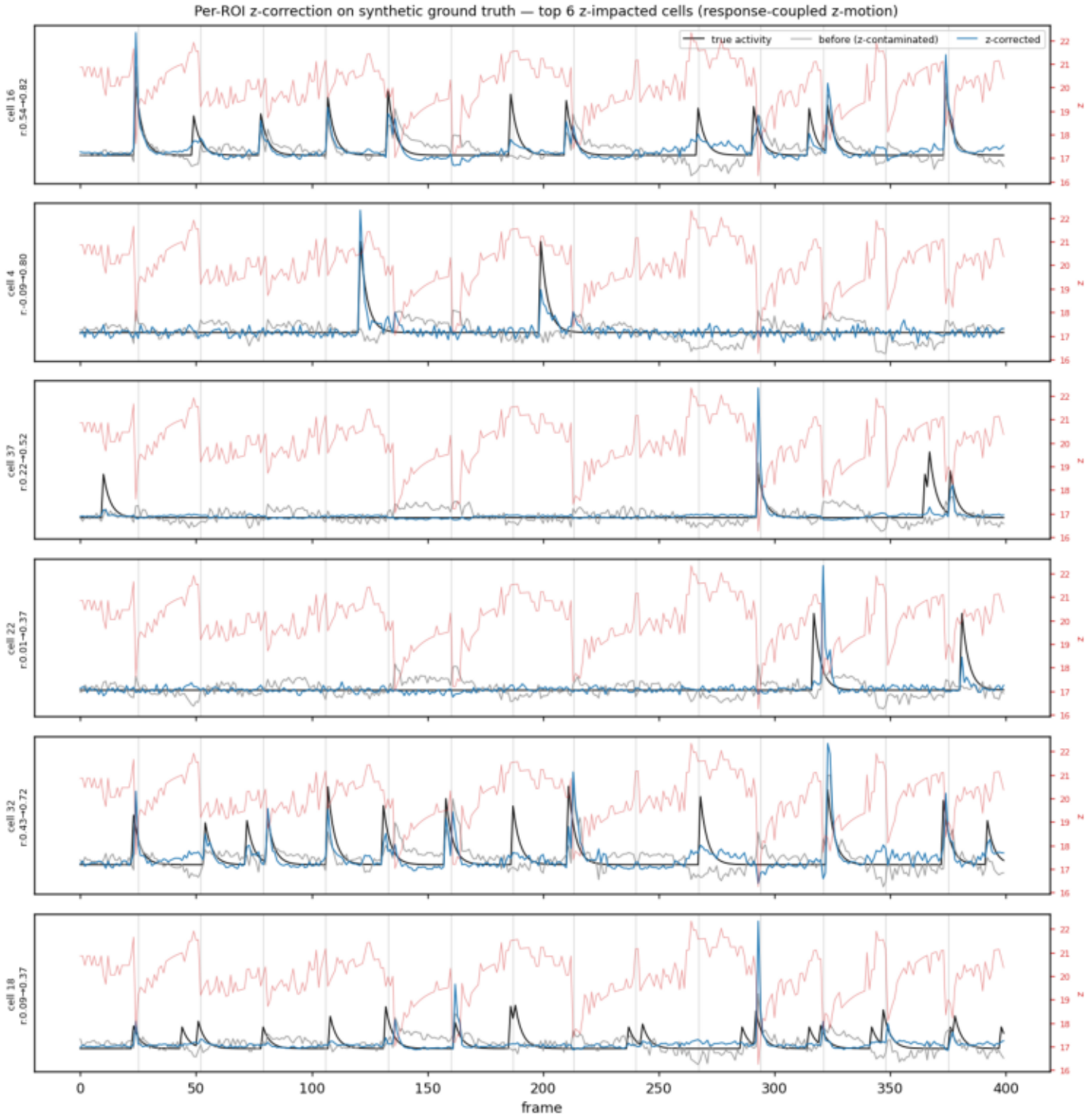
- per-pixel — regress every pixel on F_anat (a-priori; over-corrects, mixes noise).
- per-patch — one scale per spatial patch (coarse).
- per-ROI — one OLS fit per cell footprint (the shipped method).

Residual z-contamination on traces: none 0.39 → per-pixel 0.32 → per-patch 0.39 → per-ROI 0.27 (lower is better). Per-ROI removes the most z from the extracted traces while preserving activity — per-pixel looks lowest pixel-wise but over-corrects, so the ROI-mean trace keeps more residual z.



8. Correction — recovery on synthetic ground truth

On real data the benefit is hard to see (no ground truth). Injecting known activity into the real z-stack lets us quantify recovery. The z-motion is modelled on the real protocol: fast dynamics (from the real estimated zpos) plus downward motion onsets clustered around stimulus responses (grey vertical lines) — so the z-artifact partly overlaps the true responses, the hardest case. For the most z-impacted cells the correction removes the z-jitter (grey→blue) and recovers the true transients (black).



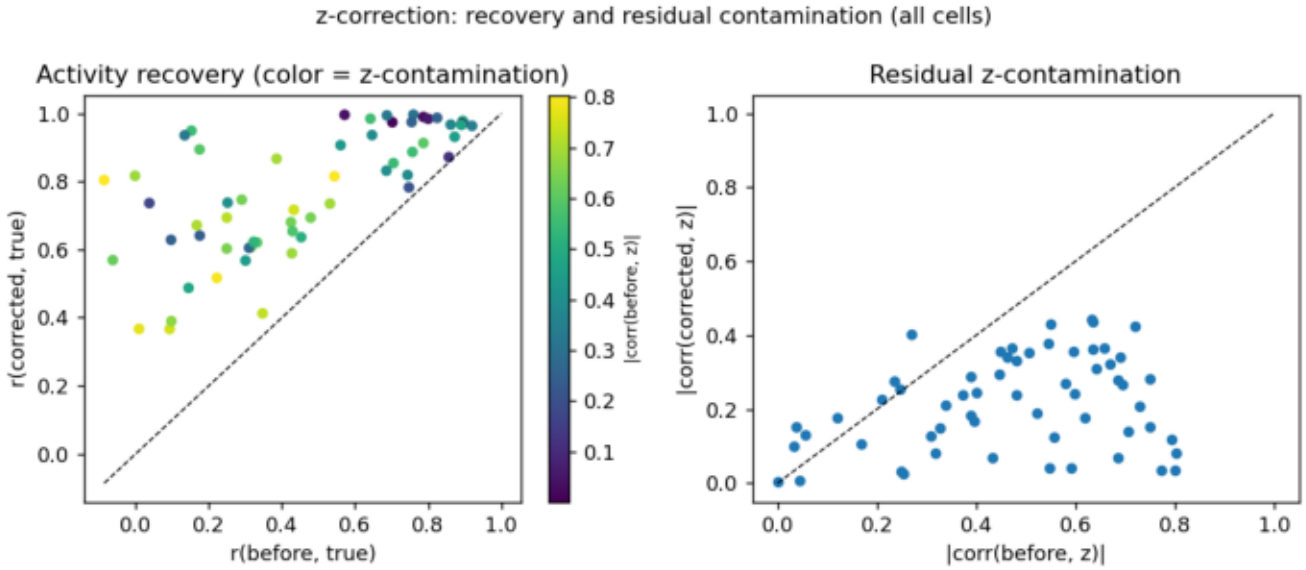
8. Correction on synthetic ground truth — recovery

All cells: recovery r 0.47 → 0.78 | residual z-contamination 0.47 → 0.22

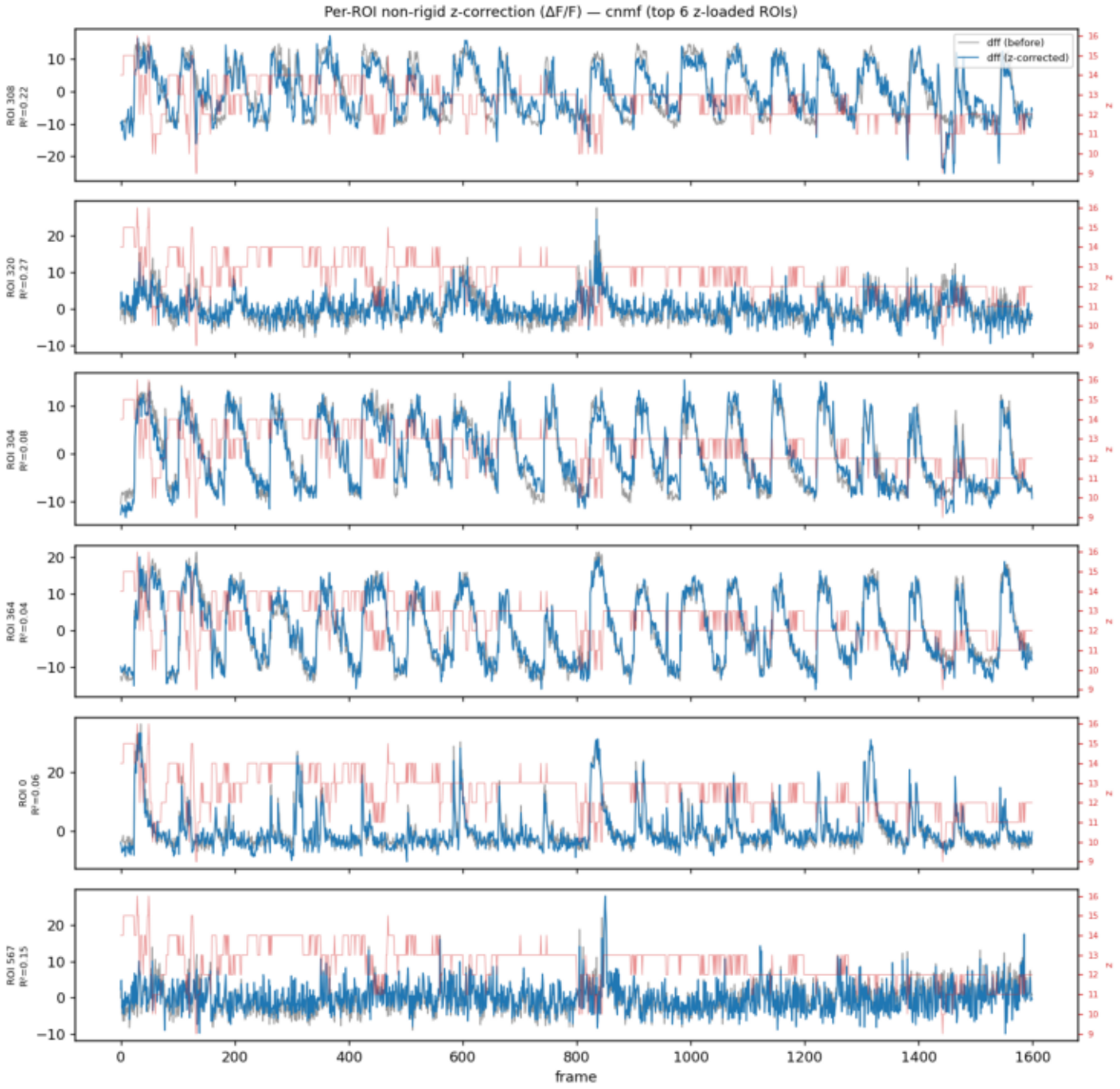
Top z-impacted (10): recovery r 0.24 → 0.62 | residual z 0.75 → 0.17

Benefit: the most z-impacted cells more than double their activity recovery and shed ~85% of z-contamination. Clean cells are left essentially unchanged.

Limit: recovery plateaus below 1.0 — z-defocus attenuates the true signal, which additive subtraction cannot restore.

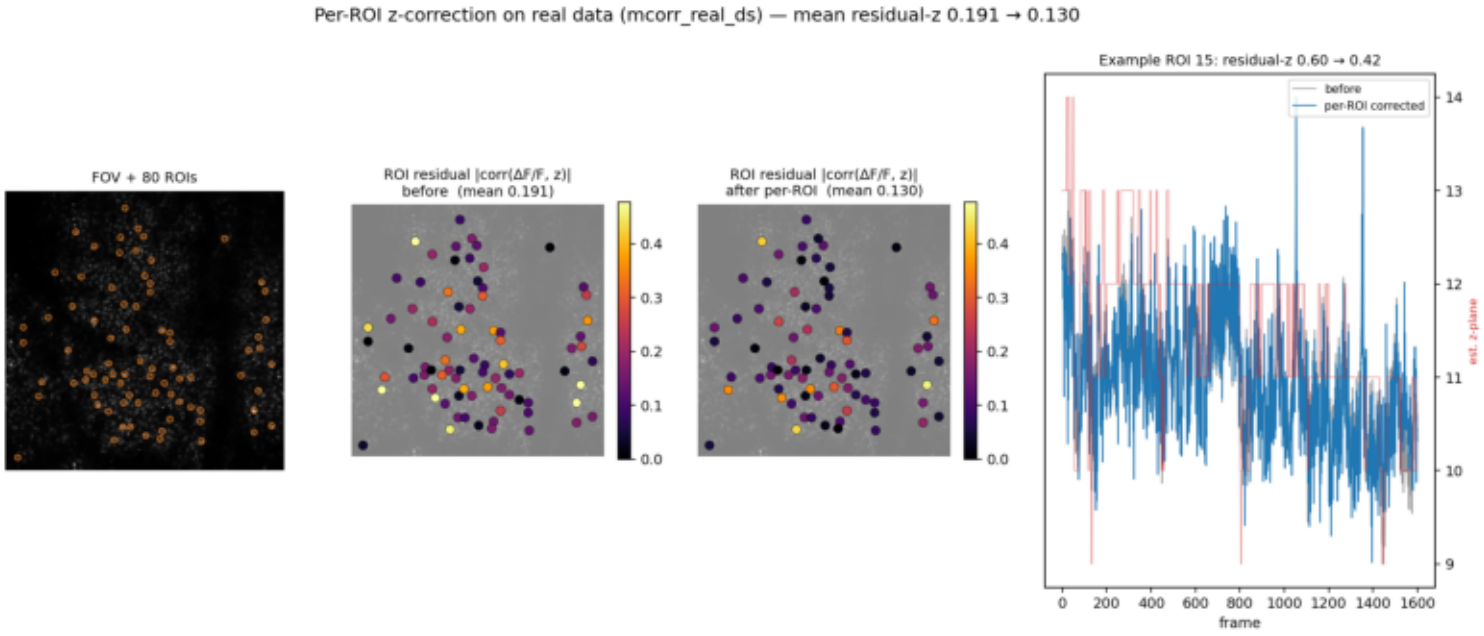


9. Per-ROI correction on real data (CNMF traces)



10. Per-ROI correction on real data (spatial)

The shipped per-ROI correction on a real motion-corrected recording (no ground truth). Left: FOV with extracted ROIs. Middle: per-ROI residual z-contamination |corr(ΔF/F, z)| before vs after correction (brighter = more z-contaminated). Right: an example ROI trace (raw vs corrected) with the estimated z overlaid — the correction removes z-locked fluctuations from the trace. Mean ROI residual-z 0.19 → 0.13 across 80 ROIs.



11. Conclusions & recommendations

What works, and what to do next

Pipeline:

• The z-motion chain is validated end-to-end on real data, both CNMF and Suite2p backends. 7 latent bugs were fixed in the per-ROI path.

z-position estimation:

• Synthetic ground truth: phase correlation (Suite2p) is most accurate (RMSE ~0.7 vs ~1.7 planes for Pearson).

• BUT on real NoRMCorre-corrected movies this does not transfer: phasecorr is much noisier — 94 vs 18 spurious ≥2-plane jumps on the 3-session concat (5×), and a deep-plane bias in single sessions — and edge-cropping collapses Pearson to plane 0. Since z-pos runs after mcorr, phasecorr is not usable there.

• The robust improvement is Pearson + temporal median smoothing of zpos: same drift as legacy but 44 vs 290 spurious ≥2-plane jumps on the real single run.

• Recommendation: keep Pearson (now the default in compute_zcorrel) with temporal smoothing; do NOT adopt phasecorr/edge-crop for production despite their synthetic accuracy. Revisit phasecorr only for raw/lightly-registered data.

F_anat & correction:

• Per-ROI fit-and-subtract removes the most z-contamination at every drift amplitude; per-pixel/per-patch trail. Recovery is capped by defocus attenuation.

• Sub-plane interpolation of the z-peak gives negligible gain (global or F_anat) — not worth adding.